

Lab 5, task #2 (Monday 25.05.2026)

Using the **PlatEMO framework (Matlab)**, solve the following tasks.

- 1) **(max. 2 points)** Create (manually) your own version of the NSGAI algorithm (name it differently!), changing the selection type from TournamentSelection to RouletteWheelSelection and specifying only the FrontNo as the fitness function.
- 2) **(max. 2 points)** Define your own optimization problem (MOO) in the "Application Module" tab based on your previous task from Lab 3: only Lab 3 pt. 1).
- 3) **(max. 3.5 points)** In the "Experiment Module" tab, configure an experiment comparing the solution of the tasks:
 - a) your own MOO problem, saved in a file, from point 2),
 - b) ZDT 3-4,
 - c) DLTZ 3-4,
 - d) WFG 3-4,using the methods:
 - i. your modified NSGAI method from point 1),
 - ii. original NSGAI,
 - iii. MOEA/D.

Make sure that the parameters for the tested methods are the same. As part of the experiment, observe the values of the GD, IGD and HV metrics, set the "**signed-rank test**" statistical analysis and the average values with standard deviations. Configure at least 20 experiment repetitions (number of runs). Save the results (for each metric) to a file (e.g. in TeX table format).

- 4) **(max. 1.5 points)** Prepare a report **presenting numerically (in tables)** and analyzing the results of the experiment from point 3).